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Assignment 1: Systematic Paper Analysis

Implicit Neural Representations with Periodic Activation Functions

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GROUP ID09 — TOPIC-WISE WORK DIVISION

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1 Introduction

Signal is data that varies over time, mathematically expressed as functions. Traditional methods store data explicitly — for example, images are stored as grids of pixels, which is a discrete representation. A continuous signal, by contrast, can be evaluated at any coordinate, not just fixed pixel positions. A function is called differentiable if its derivative (rate of change) can be computed. For an image brightness function $f(x, y)$, the derivative reveals edges and curvature.

Implicit Neural Representations (INRs) define a signal implicitly through the network weights: instead of storing pixels, a neural network computes the signal value at any input coordinate using its learned parameters $f(x, y; \theta)$. This approach offers infinite resolution, compact storage, and smooth derivatives. However, most implicit networks use the ReLU activation, which struggles with high-frequency signals such as fine textures, hair, or grass. Because ReLU is piecewise linear, its second derivative is zero everywhere, making it impossible to accurately represent curvature or satisfy physics equations that involve higher-order derivatives.

Periodic networks use a sinusoidal activation $f(x) = \sin(Wx + b)$, which naturally captures oscillations at many frequencies, produces well-behaved smooth derivatives, and can compute gradients, Laplacians, and curvature directly. The Sinusoidal Representation Network (SIREN) [1] builds on this idea and proposes a principled weight initialisation scheme to keep activations stable across layers. The authors sample weights from $\mathcal{U}(-\sqrt{6/n}, \sqrt{6/n})$ and scale the first layer by a frequency factor ω_0 , ensuring that neither vanishing nor exploding gradients occur during training. SIREN is then demonstrated across images, audio, video, 3D shapes, and partial differential equations. The study also combines SIREN with hypernetworks [16] to learn a prior over entire distributions of signals — for example, representing all possible face images rather than one at a time.

2 Paper Summary

2.1 Research Problem Addressed

Existing implicit neural representations built on ReLU or smooth non-periodic activations fail to capture fine-grained high-frequency details such as sharp textures and complex boundaries. More critically, they cannot represent higher-order derivatives accurately, which is essential for solving physics-based equations like Poisson, Helmholtz, and Eikonal equations. Even dense supervision does not fix these limitations when the activation function itself cannot represent rapid oscillations.

2.2 Motivation and Real-World Importance

Real-world signals — medical scans, audio waveforms, seismic data, 3D geometry — are continuous and derivative-dependent. Discrete grid-based representations consume large memory, have limited resolution, and cannot enforce mathematical constraints. A continuous neural representation that models both signal values and their derivatives opens the door to compact storage, resolution-free inference, and physics-based learning without explicit numerical solvers.

2.3 Narrow Target Task

The paper proposes a neural network architecture that can: (1) represent complex continuous signals at arbitrary resolution, (2) compute accurate spatial and temporal derivatives of any order, and (3) solve differential equations by supervising only on derivative constraints rather than direct signal values.

2.4 Literature Context: Prior Methods Referenced

Standard INRs use ReLU MLPs [2, 3], which are piecewise linear and therefore produce zero second derivatives. Smooth activations like tanh or softplus improve derivative quality but still fail on high-frequency content. Positional encoding methods [27] and NeRF [4] apply fixed Fourier features to inputs before passing them into a ReLU network, which helps with frequency but loses derivative coherence. Signed distance function networks [5, 6] represent 3D shapes implicitly but rely on ReLU and cannot enforce Eikonal constraints cleanly. Scene Representation Networks [7] use continuous scene representations but with smooth non-periodic activations. None of these methods unify signal fitting and derivative-based supervision under one architecture.

2.5 How the Proposed Approach Differs from Earlier Work

SIREN replaces all activations with sine functions, giving the network a closed-form derivative that is itself a SIREN. This means training on gradients or Laplacians does not require any special treatment — the loss is simply applied to the network’s analytical derivative. The proposed initialisation scheme [24, 25] ensures stable training in deep periodic networks, which prior work had not addressed. The breadth of applications — from audio to PDEs — in a single unified architecture is also new.

3 Methodology Analysis

Sitzmann et al. [1] present Sinusoidal Representation Networks (SIREN) to address the drawbacks of conventional INRs. SIREN uses sine wave activations to ensure derivatives are well-behaved and representational capacity is greater than ReLU-based architectures, which struggle with high-frequency detail and higher-

order derivatives.

3.1 Model Architecture

SIREN is a Multi-Layer Perceptron (MLP) that maps input coordinates directly to signal values. Unlike classical architectures that use non-periodic activations like ReLU, SIREN uses a periodic sine activation. The full network mapping for an L -layer SIREN is:

$$\Phi(\mathbf{x}) = \mathbf{W}_L(\phi_{L-1} \circ \phi_{L-2} \circ \dots \circ \phi_0)(\mathbf{x}) + \mathbf{b}_L \quad (1)$$

where each intermediate layer ϕ_i is defined as:

$$\phi_i(\mathbf{x}_i) = \sin(\omega_0 \cdot (\mathbf{W}_i \mathbf{x}_i + \mathbf{b}_i)) \quad (2)$$

3.1.1 Role of the Sine Activation

The sine activation is mathematically selected because its derivatives are also sine waves, making the architecture self-similar across all derivative orders. Unlike ReLU, which has a zero second derivative (zero Laplacian), SIREN can model curvature and solve Boundary Value Problems (BVPs) with derivative-based constraints such as the Eikonal equation $|\nabla \Phi| = 1$ [1, 20].

3.1.2 Frequency Scaling ω_0

The hyperparameter ω_0 scales input coordinates so the network captures high-frequency spatial or temporal data from the start. Without it, normalized inputs in $[-1, 1]$ remain in the near-linear region of sine around the origin, producing an effectively linear model. The value $\omega_0 = 30$ is empirically determined as the optimal balance between fine-detail representation and gradient stability [27].

3.1.3 Theoretical Stability and Initialisation

The authors derive a weight initialisation scheme using uniform distributions to eliminate vanishing gradients in periodic networks [24, 25]. Weights are sampled from $\mathcal{U}(-\sqrt{6/n}, \sqrt{6/n})$, keeping activation distributions constant throughout network depth. This ensures steady gradient flow and allows SIREN to converge in regimes where conventional initialisations fail.

3.2 Dataset(s) Used

To demonstrate universality, SIREN is tested on 1D audio (Bach cello suite, speech), 2D images (BSDS500 [19]), and 3D point clouds (Stanford Thai Statue) via Signed Distance Functions [2]. Temporal consistency is tested on video sequences and complex physics via the Helmholtz equation and seismic wavefields [21, 22]. Large-scale datasets (CelebA [18] and BSDS500) are used for gradient-supervised reconstruction.

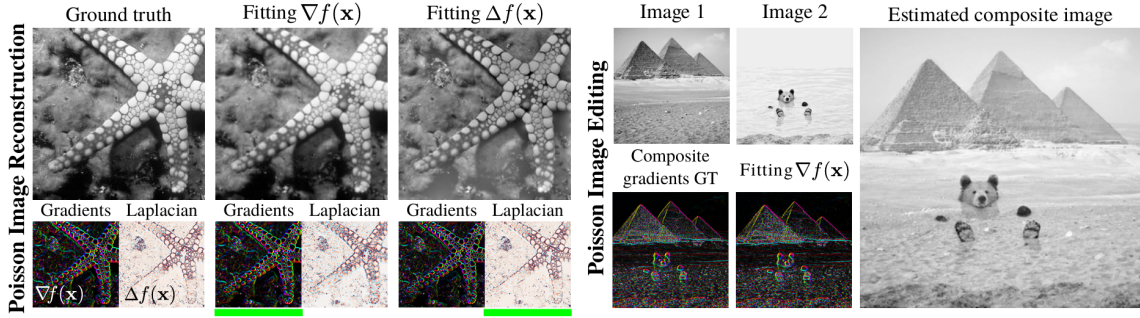


Figure 1: Activation distributions in a SIREN. The principled initialisation keeps signal distribution constant across deep layers.

3.3 Training Procedure

SIREN uses a coordinate supervision strategy, mapping input coordinates to signal values via Mean Squared Error (MSE) loss for reconstruction, or derivative-based supervision for tasks like the Eikonal equation [1]. Optimisation uses Adam at learning rate 1×10^{-4} . The framework optionally incorporates Hypernetworks or Conditional Neural Processes to generalise across signal distributions [16, 17]. For physics-specific constraints, ADMM ensures mathematical consistency [26].

3.4 Evaluation Metrics

Reconstruction quality is measured by Peak Signal-to-Noise Ratio (PSNR) for images and video, and Mean Squared Error (MSE) for 1D audio. For 3D geometry, Intersection over Union (IoU) and Chamfer Distance evaluate accuracy of learned SDF models against ground-truth surfaces [1].

3.5 Baselines Used for Comparison

SIREN is compared against standard ReLU MLPs, Tanh, and Softplus networks with conventional initialisations [24, 25]. Most importantly, it is compared to Fourier Feature mappings and Positional Encoding [27], the leading approaches for adding high-frequency detail to implicit representations. On 3D tasks, DeepSDF [2] serves as the specialised baseline.

4 Experimental Setup and Results Analysis

4.1 Main Experimental Results

SIREN produces significantly more faithful reconstructions than prior systems on images, recovering sharp edges and fine textures that ReLU-based models smear. On 1D audio it preserves high-frequency harmonics that ReLU models completely miss. Most importantly, it successfully handles complex Boundary Value Problems: signal reconstruction from gradients via the Poisson equation, and 3D shape representation via Signed Distance Functions [1, 20].

4.2 Performance Comparison with Baselines

SIREN outperforms ReLU, Tanh, and Softplus MLPs [24, 25] across all tasks. Compared to Fourier Features [27], which produce blob artifacts in the Laplacian ($\Delta\Phi$), SIREN yields clean analytical derivatives suitable for equation modelling such as Helmholtz [21]. It achieves higher quantitative PSNR across all datasets with fewer parameters than positional encoding baselines.

4.3 Ablation Studies

- **Initialisation impact:** Conventional Xavier [24] and He [25] schemes introduce vanishing gradients in periodic networks; the proposed uniform initialisation is necessary for stable convergence.
- **Frequency scaling ω_0 :** Without first-layer frequency scaling, the architecture degrades to a low-capacity linear model that cannot represent fine detail.
- **Scalability:** Combining Hypernetworks [16] and Conditional Neural Processes [17] extends SIREN to generalise across full data distributions such as CelebA [18].

4.4 Error Analysis

SIREN is sensitive to the hyperparameter ω_0 : an incorrect choice can produce non-convex loss landscapes and local minima. In inverse problems such as full-waveform inversion (FWI), sparse data can cause high-frequency artifacts and ghosting. These are reduced through ADMM, which imposes physical constraints during optimisation [22, 26].

5 Critical Analysis

SIREN achieves strong results across many tasks, but a careful reading of the paper and supplementary material reveals real weaknesses in the method, its evaluation, and its reproducibility.

5.1 Limitations of the Method

No principled way to choose ω_0 . The entire method depends on a frequency scaling parameter $\omega_0 = 30$ applied to the first layer. This value was chosen by trial and error, not derived from theory. There is no formula to determine the right ω_0 for a new signal type. The supplement also quietly applies a different scale for audio by changing the input domain from $[-1, 1]$ to $[-100, 100]$, equivalent to multiplying ω_0 by 100. This inconsistency shows that $\omega_0 = 30$ does not even generalise across the paper’s own experiments. NeurIPS Reviewer 3 raised this concern directly; the authors tested deviation only on one image task and found robustness up to 20% — insufficient evidence across the full task suite.

Training is very slow and done per signal. SIREN trains one separate network for each signal from scratch. Training times from the supplementary material are shown in Table 1. Each row represents fitting a *single* signal. Scaling to even 100 images would require hundreds of GPU-hours, making SIREN impractical for any real dataset.

Task	Iterations	Time
Image fitting (Poisson)	10,000	~90 min
3D shape (SDF)	50,000	~6 h
Helmholtz equation	50,000	~8 h
Full-waveform inversion	90,000	~27 h
Wave equation	100,000	~25 h

Table 1: Per-signal training times (SIREN supplementary material).

Deeper networks produce unstable training. Using 15 layers instead of 5 improves accuracy on the Poisson task but introduces visible spikes in the loss curve. The authors say the network recovers, but no experiments go beyond 15 layers, so it is unknown whether this instability worsens with more depth or harder tasks.

5.2 Dataset and Evaluation Limitations

The paper tests SIREN on six modalities, which appears thorough. In reality, each modality is tested on very few examples:

- **Images:** A single starfish photo from BSDS500 [19] (256×256) is used for almost every image experiment. No average PSNR over a set of images is ever reported.
- **Audio:** Two clean studio clips only — 7 seconds of Bach and a 12-second counting recording. Noisy or compressed audio is never tested.
- **Video:** One cat video (300 frames). A second video appears only in the supplement. No standard benchmark is used.
- **3D shapes:** Two objects only. Quantitative Chamfer distances were absent from

the main paper and added only after Reviewer 2 requested them.

- **PDEs:** All domains are 2D and synthetic. No 3D problem is tested anywhere in the paper or supplement.

Figure 2 illustrates the audio result — SIREN perfectly recovers the Bach waveform while ReLU produces a nearly flat output. This looks compelling but rests on just two recordings.

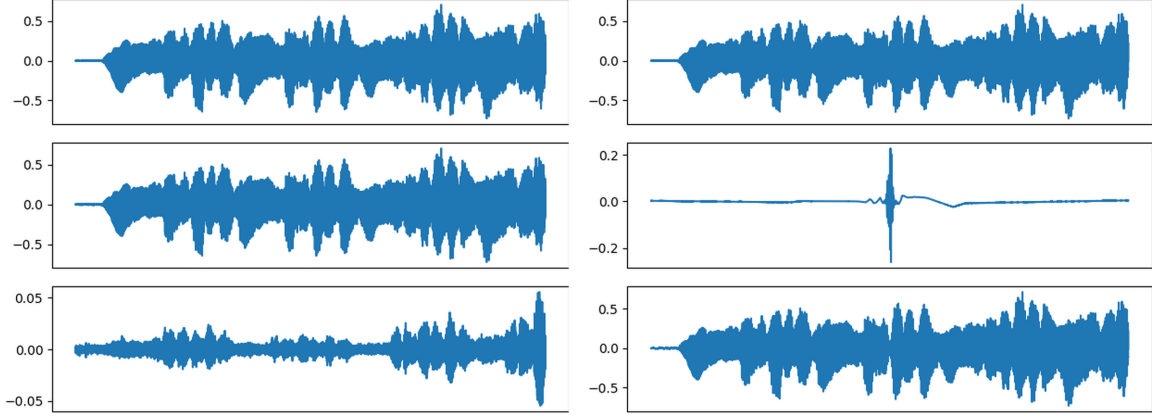


Figure 2: Audio waveform fitting on Bach’s Cello Suite. **Left (SIREN):** ground truth, predicted output, and error — all closely matched. **Right (ReLU):** predicted output is nearly flat — complete failure. Only two audio clips were tested in the entire paper. (*Supplementary material, Figure 10.*)

5.3 Reproducibility Concerns

Loss weights are task-specific and unjustified. Each experiment combines multiple loss terms with hand-tuned constants. For the SDF task the supplement gives $\lambda_1 = 50$, $\lambda_2 = 3000$, $\lambda_3 = 100$; for FWI the weight equals the batch size. No sensitivity analysis is provided for any of these values, making it unclear whether results would hold under different settings.

Complex experiments reported from a single run. Audio and inpainting results include mean and standard deviation over 10 runs. The harder experiments — SDF, Helmholtz, FWI, wave equation — are each reported from one run only, with no variance. These are precisely the cases where reliability matters most.

5.4 Scalability Issues

One network per signal. Every experiment except CelebA requires a separate network trained from scratch. The proposed hypernetwork [16] is meant to fix this but only works at 32×32 resolution, far below practical use.

PDE experiments are 2D only. All PDE domains are 2D and synthetic. Real problems in seismic imaging, fluid dynamics, and electromagnetics are 3D and time-

dependent. Whether SIREN can scale to these settings is completely unknown.

6 Research Opportunities

Based on the limitations above, we propose three extensions to the SIREN framework, focusing on new datasets, model modifications, and efficiency improvements.

6.1 Extension 1: Evaluation on a Larger and More Diverse Image Dataset

Motivation. The paper draws all image conclusions from a single photograph. Whether SIREN consistently outperforms baselines across different image categories — textures, faces, medical scans, satellite imagery — is unknown. This extension addresses the dataset limitation directly.

Proposed extension. Evaluate SIREN and its three strongest baselines (ReLU, Tanh, ReLU-PE) on the full BSDS500 benchmark [19] (500 natural images) and additionally on a medical imaging dataset such as ChestX-ray14 or Brain MRI, which have very different frequency characteristics from natural photographs. Report mean and standard deviation of PSNR, gradient-PSNR, and Laplacian-PSNR across all images for each modality, giving the first statistically meaningful comparison.

Expected outcome. SIREN should maintain its advantage on high-frequency natural images but show a smaller gap on smooth, low-frequency medical images where ReLU-PE is already competitive. This defines the operating boundary of SIREN and guides practitioners on when the method is worth its computational cost.

6.2 Extension 2: Adaptive Frequency Scaling as a Model Modification

Motivation. The fixed $\omega_0 = 30$ is the most fragile part of SIREN. A straightforward model modification — making ω_0 a learnable parameter per layer — could remove this bottleneck and make the method much easier to apply to new tasks without manual tuning.

Proposed extension. Replace the fixed ω_0 with a learnable scalar per layer, initialised from the dominant frequency of the target signal computed via a Fast Fourier Transform (FFT) before training begins. Each layer’s frequency scale is updated by gradient descent alongside the network weights. This adds only one scalar parameter per hidden layer (five extra parameters for a 5-layer SIREN) with no other architectural changes. Compare the modified SIREN against the original on images, audio, SDF, and PDE tasks using the same training budgets as the paper.

Expected outcome. Learnable ω_0 should remove manual tuning, improve performance on signals whose dominant frequency differs significantly from 30 (e.g., high-rate audio, medical images, large-scale 3D geometry), and resolve the inconsistency where audio silently requires a $100\times$ larger input scale.

6.3 Extension 3: Knowledge Distillation for Efficient SIREN Compression

Motivation. SIREN networks are slow to train and large relative to the signals they represent. An efficiency improvement via knowledge distillation could make SIREN practical for deployment, particularly for audio and image compression applications where a compact, fast model is essential.

Proposed extension. Train a full-size SIREN (5 layers, 256 hidden units — the teacher) on a target signal. Then train a smaller student SIREN (3 layers, 64 units) to match both the signal output values and the intermediate activations of the teacher at sampled coordinates. Evaluate compression ratio (parameter count), reconstruction PSNR, and training time on BSDS500 [19] for images and on the paper’s audio clips. Compare against classical compression baselines (JPEG for images, MP3 for audio).

Expected outcome. A 3-layer, 64-unit student has roughly $12\times$ fewer parameters than the teacher. We expect a modest PSNR drop of 1–3 dB at this ratio, which would make SIREN competitive with classical codecs on short clips and open a practical path toward neural signal compression.

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